DFW Housing Market Analysis

Methodology Document

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Project Type: End-to-End Data Analytics Portfolio Project
Analysis Period: May 2016 – May 2026
Tools Used: Google BigQuery, SQL, Tableau, Microsoft Excel, GitHub

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1. Project Overview

Purpose

The purpose of this project was to analyze residential housing market trends across the Dallas–Fort Worth (DFW) metropolitan area and identify cities that offer the best balance of affordability, historical appreciation, and buyer opportunity.

The project demonstrates the complete data analytics lifecycle, including data collection, preparation, analysis, visualization, and business reporting. The final deliverables provide actionable insights for prospective homebuyers and illustrate the practical application of modern data analytics tools.

Business Question

Which Dallas–Fort Worth cities provide the best balance of affordability, historical appreciation, and buyer opportunity?

Answering this question required evaluating multiple market indicators rather than relying solely on current home prices.

Project Objectives

The project was designed to:

- Analyze current home values across DFW cities.
- Measure ten-year home value appreciation.
- Compare housing trends at both the city and county levels.
- Identify markets that outperform regional averages.

- Develop a Buyer Opportunity Index by combining affordability and appreciation rankings.
- Present findings through interactive business intelligence dashboards.

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2. Data Source

Primary Data Source

The analysis was based on the Zillow Home Value Index (ZHVI), which provides monthly estimates of typical home values across U.S. housing markets.

Analysis Scope

- Geographic Area: Dallas–Fort Worth Metropolitan Area
- Geographic Levels: City and County
- Time Period: May 2016 through May 2026

Analytical Datasets

Five datasets were created to support the analysis:

Latest Home Values

Current estimated home values for each city.

City Appreciation

Ten-year percentage appreciation for each city.

County Appreciation

Average appreciation calculated at the county level.

Buyer Opportunity Index

A composite dataset combining affordability rankings and appreciation rankings into a single Buyer Opportunity Score.

County Performance

10-year analysis of counties within the DFW Metroplex.

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3. Data Preparation

Data Import

The Zillow dataset was imported into Google BigQuery to provide a scalable environment for SQL analysis.

Data Review

Before analysis, the dataset was reviewed to ensure:

- Required fields were present.
- Records covered the full analysis period.
- Geographic information was consistent.
- Data was organized for efficient querying.

Dataset Organization

Separate analytical datasets were created to support different business questions.

This approach reduced query complexity, improved dashboard performance, and made each dataset easier to maintain.

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4. SQL Analysis

Google BigQuery was used to transform raw housing data into business-ready analytical datasets.

Current Market Analysis

SQL queries calculated:

- Average home value
- Highest home value
- Lowest home value
- Cities above the regional average

These metrics formed the foundation of the Current Market Snapshot dashboard.

Home Value Appreciation Analysis

Historical home values were compared across the ten-year analysis period to calculate appreciation percentages for each city.

Additional SQL queries summarized appreciation at the county level, allowing comparisons across broader geographic regions.

Buyer Opportunity Index

To evaluate markets using multiple factors, a Buyer Opportunity Index was developed.

The methodology combined:

- Current home value rankings (affordability)
- Historical appreciation rankings (growth potential)

Each city received a Buyer Opportunity Score based on its relative performance across both measures.

Cities were then ranked from highest to lowest opportunity.

County Performance Model

- **Home Value Rank:** County ranking based on average home value.
- **Appreciation Rank:** County ranking based on 10-year home value appreciation.
- **Buyer Opportunity Rank:** County ranking based on the Buyer Opportunity Score.
- **County Performance Score:** The average of the three component ranks.
- **Overall Rank:** The final ranking based on the composite County Performance Score.

A lower County Performance Score indicates stronger overall performance under this model. This score is a comparative measure, not a prediction of future appreciation or investment returns.

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5. Data Visualization

Tableau was used to transform analytical datasets into interactive dashboards designed for executive decision-making.

Three dashboards were created.

Dashboard 1: Current Market Snapshot

Purpose:

Provide an overview of current housing prices across the DFW region.

Included:

- Average Home Value KPI
- Highest Home Value KPI
- Lowest Home Value KPI
- Cities Above Average KPI
- Average Home Value by County
- Highest Value Cities
- Lowest Value Cities
- Geographic Bubble Map

Dashboard 2: Home Value Appreciation

Purpose:

Analyze historical market growth across cities and counties.

Included:

- Average Appreciation KPI
- Highest Appreciation KPI
- Lowest Appreciation KPI
- Cities Above Average KPI
- Highest Appreciating Cities
- Lowest Appreciating Cities
- Average Appreciation by County
- Geographic Bubble Map

Interactive dashboard actions allow users to filter all visualizations by selecting a county or city.

Dashboard 3: Buyer Opportunity Index

Purpose:

Identify cities offering the strongest balance of affordability and appreciation.

Included:

- Highest Opportunity Score KPI
- Average Opportunity Score KPI
- Lowest Opportunity Score KPI
- Cities Above Average Opportunity Score KPI
- Opportunity Score by City
- Top Buyer Opportunity Cities
- Lowest Opportunity Cities
- Geographic Bubble Map

This dashboard provides a consolidated view of buyer-focused market opportunities.

Dashboard 4: Executive Market Performance

Purpose: Compare county-level housing market performance using average home values, historical appreciation, buyer opportunity scores, and a composite performance ranking.

Key components:

- Highest Performing County KPI
- Highest Average Home Value KPI
- Highest Appreciation KPI
- Best Buyer Opportunity County KPI
- County Performance Score chart
- County Performance Map

- Average Home Value by County
- 10-Year Appreciation by County
- Average Buyer Opportunity Score by County
- Executive Key Insights

The findings should reflect your published dashboard: Collin County leads overall, Tarrant County has the highest average home value, Dallas County has the highest appreciation, and Rockwall County has the highest Buyer Opportunity Score.

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6. Dashboard Design Standards

Consistency was maintained across all dashboards to improve usability and create a professional user experience.

Dashboard Size

- 1400 × 900 pixels

Color Palette

- Blue for standard visualizations
- Orange highlight for the primary point of emphasis
- Gray subtitles
- White background

Only one data element is highlighted in orange within each visualization to direct user attention without overwhelming the dashboard.

User Interaction

Interactive dashboard actions were implemented to enable cross-filtering between charts and maps.

This allows users to explore related data without navigating to separate dashboards.

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7. Quality Assurance

Several validation steps were completed throughout the project.

SQL Validation

Query results were reviewed to verify:

- Correct calculations
- Accurate rankings
- Consistent aggregation
- Logical outputs

Dashboard Validation

Visualizations were reviewed to ensure:

- KPI values matched SQL outputs.
- Chart labels were accurate.
- Maps displayed the correct locations.
- Filters functioned as intended.
- Formatting remained consistent across dashboards.

Issue Resolution

During development, dashboard containers resized when filters reduced the number of displayed marks.

Multiple layout strategies were evaluated. The final design was retained because the resizing did not impact readability, usability, or the professional appearance of the dashboards.

Additionally, an issue during development caused one worksheet in the Buyer Opportunity dashboard to display county-level results instead of city-level results. The worksheet was restored by replacing the county dimension with the city dimension, reapplying the correct highlighting logic, and updating the worksheet title to maintain consistency with the project's business objectives.

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8. Project Deliverables

The completed project includes:

- SQL analysis performed in Google BigQuery
- Four analytical datasets

- Three interactive Tableau dashboards
- GitHub repository
- README documentation
- Executive Summary
- Methodology Document

Together, these deliverables provide a complete end-to-end analytics portfolio project.

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9. Skills Demonstrated

This project demonstrates proficiency in:

Data Analytics

- Data collection
- Data preparation
- Data validation
- Data transformation
- Exploratory analysis

SQL

- Data querying
- Aggregations
- Ranking
- Calculated metrics
- Business-focused analysis

Business Intelligence

- Dashboard development
- KPI design
- Geographic visualization
- Interactive filtering

- Executive reporting

Data Storytelling

- Business problem definition
- Insight generation
- Executive communication
- Actionable recommendations

Professional Documentation

- Technical methodology
- Executive reporting
- Project documentation
- Portfolio presentation

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10. Project Workflow Summary

The project followed a structured analytics workflow representative of professional business intelligence engagements.

1. Defined the business problem and project objectives.
2. Collected housing market data from the Zillow Home Value Index.
3. Imported and organized data within Google BigQuery.
4. Developed SQL queries to calculate key metrics and rankings.
5. Created four analytical datasets to support reporting.
6. Built four interactive Tableau dashboards using consistent design standards.
7. Validated calculations, visualizations, and dashboard functionality.
8. Produced executive and technical documentation to communicate findings.
9. Published a complete portfolio project demonstrating the end-to-end data analytics lifecycle.

This structured methodology ensures the analysis is transparent, reproducible, and aligned with industry best practices for business intelligence and data analytics projects.